Year 10

Financial Mathematics Test

2019

Baldivis Secondary College

Simple & Compound Interest

Name: Solutions   Time: 35 minutes   Total Mark:____/33

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning will be allocated zero marks.

Resources: 1 A4 page of notes (1 side), Calculator allowed.

Question 1	**4 marks (2, 2)**

Calculate the **amount of Simple interest** and the **total repayment** on the following.

a) $1000 is borrowed for 2 years and the interest rate is 5% pa.

$I = P \times r \times t$

$= 1000 \times 0.05 \times 2$

$= \$100$ ✓

$A = 1000 + 100 = \$1100$ ✓

b) A rate of 7.5% pa over a year on $4700.

$I = P \times r \times t$

$= 4700 \times 0.075 \times 1$

$= \$352.50$ ✓

$A = 4700 + 352.50 = \$5052.50$ ✓

Question 2	**2 marks**

$5500 is borrowed at 9.5% pa simple interest. $2090 in interest is repaid with the loan. How long did it take to repay the loan?

$I = P \times r \times t$

$2090 = 5500 \times 0.095 \times t$ ✓ correct substitution

$t = 4$ years ✓

6

Question 3	**2 marks**

At what simple interest rate will an investment of $6500 grow to a total investment of $9750 in 8 years?

$I = 9750 - 6500 = 3250$

✓ $3250 = 6500 \times r \times 8$

$r = 0.0625$

$= 6.25\%$ ✓

Question 4	**3 marks**

How much will $8000 be worth if it has been invested for 3 years with an interest rate of 2% p.a. compounding yearly? Give your answer correct to the nearest dollar.

$A = P\left(1 + \frac{r}{n}\right)^{nt}$ ✓

$= 8000\left(1 + \frac{0.02}{1}\right)^{1\times3}$

$= 8489.66$ ✓

($\frac{1}{2}$ mark for $8489)

Question 5	**6 marks (3, 3)**

A $10 000 investment is placed in a bank returning 6% p.a. compound interest. Find the value of the investment, correct to the nearest cent, after a year if interest is calculated every:

(a) six months

$A = P\left(1 + \frac{r}{n}\right)^{nt}$

$= 10000\left(1 + \frac{0.06}{2}\right)^{2\times1}$ ✓ ✓

$= \$10,609.00$ ✓

(b) month.

$A = P\left(1 + \frac{r}{n}\right)^{nt}$

$= 10000\left(1 + \frac{0.06}{12}\right)^{12\times1}$ ✓ ✓

$= \$10\,616.78$ ✓

Question 6	6 marks (4, 2)

$80 000 is invested in a bank account for 5 years at 9% per annum compound interest, (compounding yearly). Find:

(a) The amount in the account after five years. Give your answer correct to the nearest dollar.

$A = P(1 + \frac{r}{n})^{nt}$ ✓

$= 80000(1 + \frac{0.09}{1})^{1 \times 5}$ ✓

$= 123089.91$ ✓

$= \$123\,090$ (nearest \$) ✓

$P = 80000$
$t = 5$
$r = 0.09$
$n = 1$

(b) The amount of interest the $80 000 earned over the five years.

$I = A - P$ subtract ✓

$= 123090 - 80000$

$= \$43090$ ✓

Question 7	5 marks

The company Johnson and Sons purchases new machinery for $300,000 at the start of 2020. It depreciates at a rate of 15% p.a. What will the value of this machinery be on 1 July 2025?

$t = 5.5$ years ✓

$P = 300000$

$r = 0.15$

$A = P(1 - r)^t$ ✓

$= 300000(1 - 0.15)^{5.5}$ ✓ ✓

$= \$122\,722.83$ ✓

Question 8	5 marks (2, 2, 1)

Todd deposited $1,600 in an account that earns 7% simple interest every year.
His brother Adam deposited $1,550 in an account that earns 7.5% interest compounded annually.

The deposits were made on the same day, and no additional money will be deposited or withdrawn from the accounts.

Who will have the most money in their account, and how much more, at the end of 4 years?

Todd

$I = P \times r \times t$

$= 1600 \times 0.07 \times 4$

$= \$448$ ✓

$A = 1600 + 448$

$= \$2048$ ✓

Adam

$A = P\left(1 + \frac{r}{n}\right)^{nt}$

$= 1550\left(1 + \frac{0.075}{1}\right)^{1 \times 4}$ ✓

$= \$2069.98$ ✓

Adam has more ✓($\frac{1}{2}$)

$2069.98 - 2048 = \$21.98$ ✓($\frac{1}{2}$)

End of Test